ID	joint	board serial	sessions (min)	total (min)	slow-speed friction (N·m)	break-in-speed friction (N·m)	rate (%/h)	FET max (°C)	R (mΩ)	L (µH)
01	left hip roll	328934703435	20.5 + 60.2	80.6	0.422 → 0.365 (−13%)	0.419 → 0.357 (−15%)	−4.9	33.4	67.9	38.5
02	right hip roll	3272346E3435	20.6 + 1.1 + 60.2	81.8	0.377 → 0.346 (−8%)	0.377 → 0.355 (−6%)	−4.2	32.9	68.9	39.1
03	left hip yaw	325234743435	3.2 + 80.5	83.7	0.409 → 0.341 (−17%)	0.479 → 0.371 (−23%)	−6.3	32.5	68.1	37.1
04	right hip yaw	3273346A3435	3.2 + 80.5	83.7	0.330 → 0.321 (−3%)	0.340 → 0.331 (−3%)	−2.4	31.5	68.2	36.3
05	left hip pitch	326234783435	3.2 + 80.5	83.7	0.407 → 0.324 (−20%)	0.435 → 0.340 (−22%)	−5.8	31.4	67.2	37.8
06	right hip pitch	327634873435	3.2 + 80.5	83.7	0.430 → 0.387 (−10%)	0.458 → 0.412 (−10%)	−3.8	30.0	67.2	38.3
07	left knee pitch	325834543435	2.1† + 3.2 + 80.5	85.8	0.355 → 0.320 (−10%)	0.391 → 0.341 (−13%)	−6.6	32.7	68.7	37.9
08	right knee pitch	3277345A3435	3.2 + 80.5	83.7	0.057 → 0.050 (−12%)	0.057 → 0.046 (−20%)	−13.2	34.4	67.9	38.0
mean (n = 7)				83.3	0.390 ± 0.037 (9%) → 0.343 ± 0.025 (7%)	0.414 ± 0.048 (12%) → 0.358 ± 0.027 (8%)	−4.9 ± 1.5		68.0 ± 0.6	37.8 ± 0.9

Slow-speed friction: first check (cold) → last check (warm). Break-in-speed friction: first 2 min → last 2 min. Rate: last 20 min of the longest session. † Stopped by DC-bus undervoltage, no interval log. Red: friction ≈ 1/7 of the other units (reducer likely not engaged), not in the mean row.